

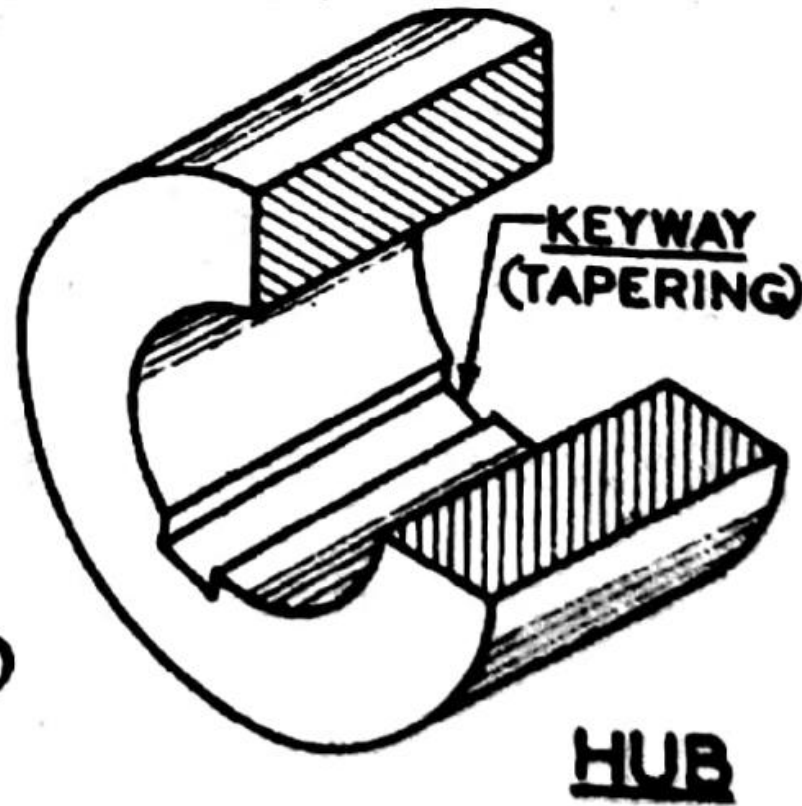
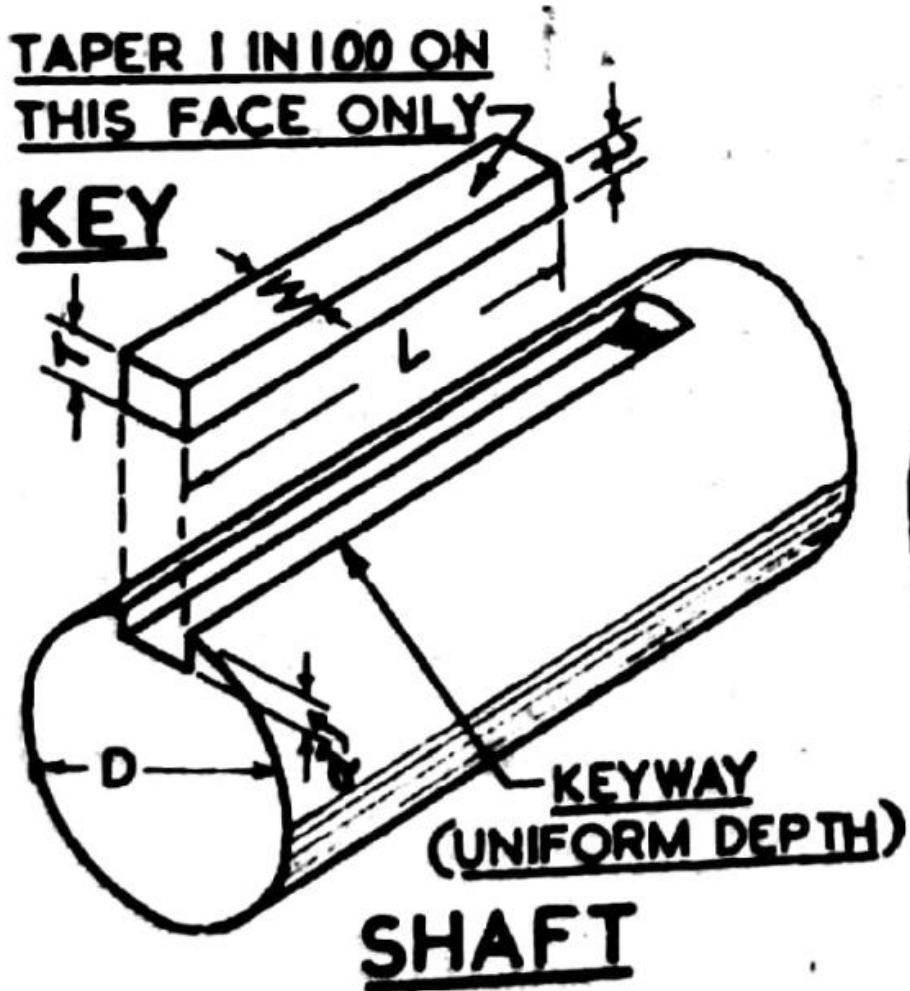
KEYS

Key is a machine component, inserted between shaft and hub of the rotor to prevent circumferential relative movement between them.

Keyway is the groove milled on the shaft and its mating part. Key is inserted in these grooves after aligning them.

TAPER 1 IN 100 ON
THIS FACE ONLY

KEY



Key and keyways

TYPES OF KEYS

Keys can be broadly classified in four categories;

saddle keys, sunk keys, parallel keys and woodruff keys.

Saddle Keys

Saddle keys are sunk only in hub and not in shaft. They are likely to slip and hence used only for light work. They transmit torque only by friction between the metallic surfaces.

Thickness , $T = D/12$ Or

Width , $W = D/4$

Width of key, $W = 0.25 D + 2 \text{ mm}$

Nominal thickness, $T = 0.08 D + 1 \text{ mm}$.

where D is the shaft diameter.

A taper of 1:100 is given on thickness.

Only frictional force is used to keep them in position. Their bottom face can be flat or hollow.

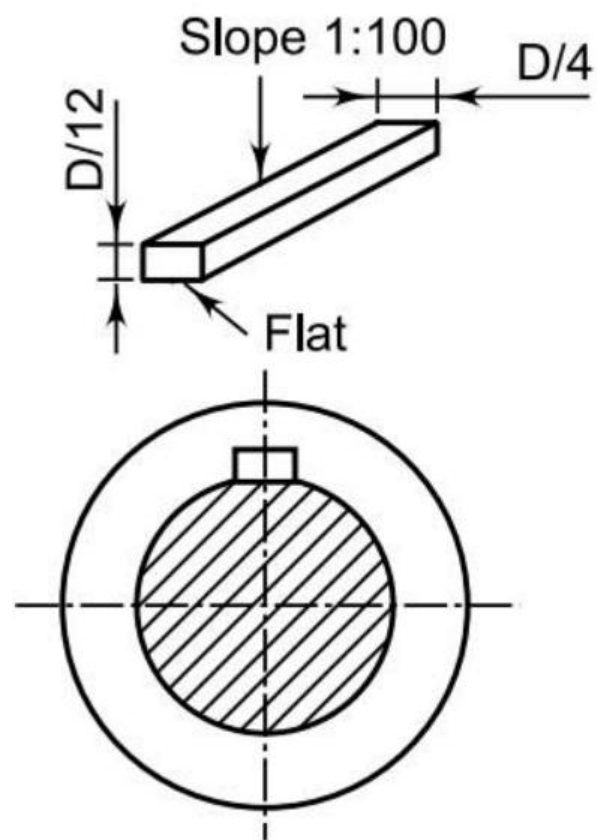
Thickness at the thicker end is called nominal thickness.

For a **flat key**, the shaft is also made flat at the position of key.

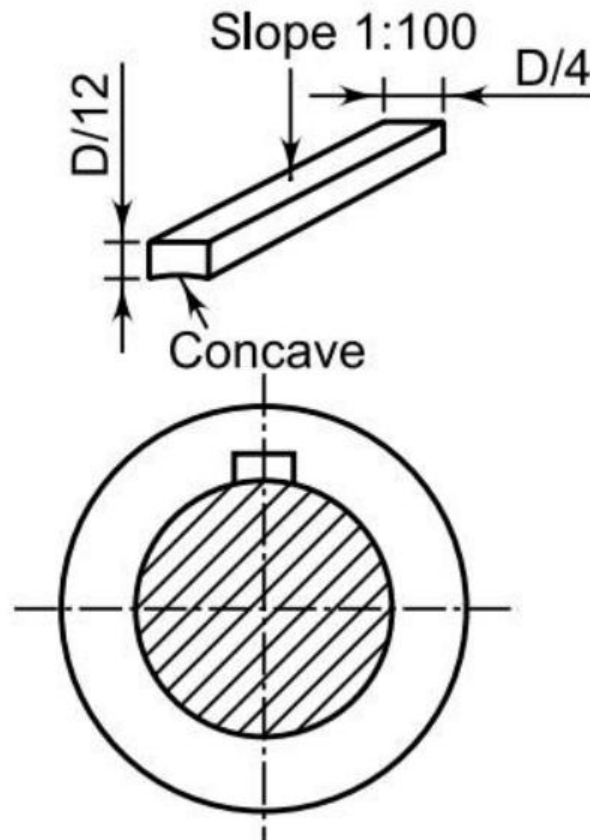
A **hollow key** (concave key) has the same curvature at the bottom as the shaft. The shaft is not modified at all.

Flat key is slightly superior to hollow key as the flat surface of the shaft resists relative motion.

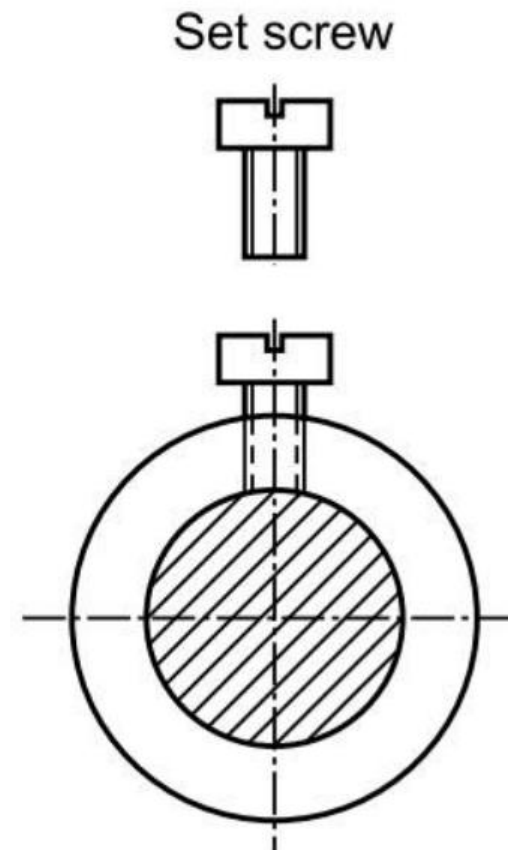
Sometimes a set screw is also used for light applications to tighten the hub on the shaft , e.g. radio knobs are tightened with a screw on its side.



(A) Flat key



(B) Hollow key



(C) Set screw

Sunk Keys

These keys are sunk half in shaft and half in hub of the rotor.

They are used for medium and heavy duty work for a positive drive till sheared off.

These are uniform in width but thickness has a taper of 1:100 along length to prevent sliding.

Length of these keys is kept from D to $1.5D$.

Sunk keys can be tapered or parallel.

Taper sunk keys

A) Square Key

Square key has square cross-section.

The thumb rule can be used as:

Width of key (W) = $0.25D$ Thickness of key (T) = $0.25D$

B) Rectangular Key

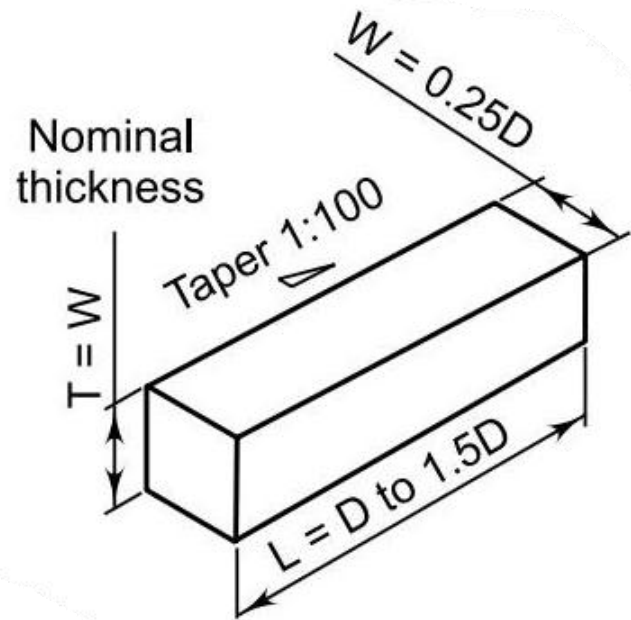
Rectangular key has rectangular cross-section and is thinner but wider than square key .

The thumb rule can be used as:

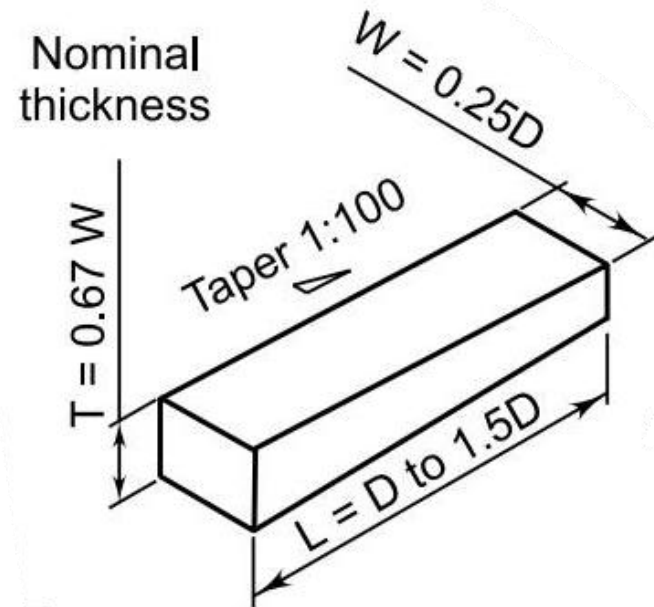
Width of key (W) = 0.25D

Thickness of key (T) = 0.67W

Width of key, $W = 0.25 D + 2 \text{ mm}$
Nominal thickness, $T = 0.2 D + 2 \text{ mm.}$



Square Key



Rectangular Key

C) Gib Head Key

A tapered sunk key can be removed by driving it out from the other exposed end. If this end cannot be accessed, then a gib is provided at one end of the key. Hammering a tapered wedge between the gib and boss helps in driving the key out.

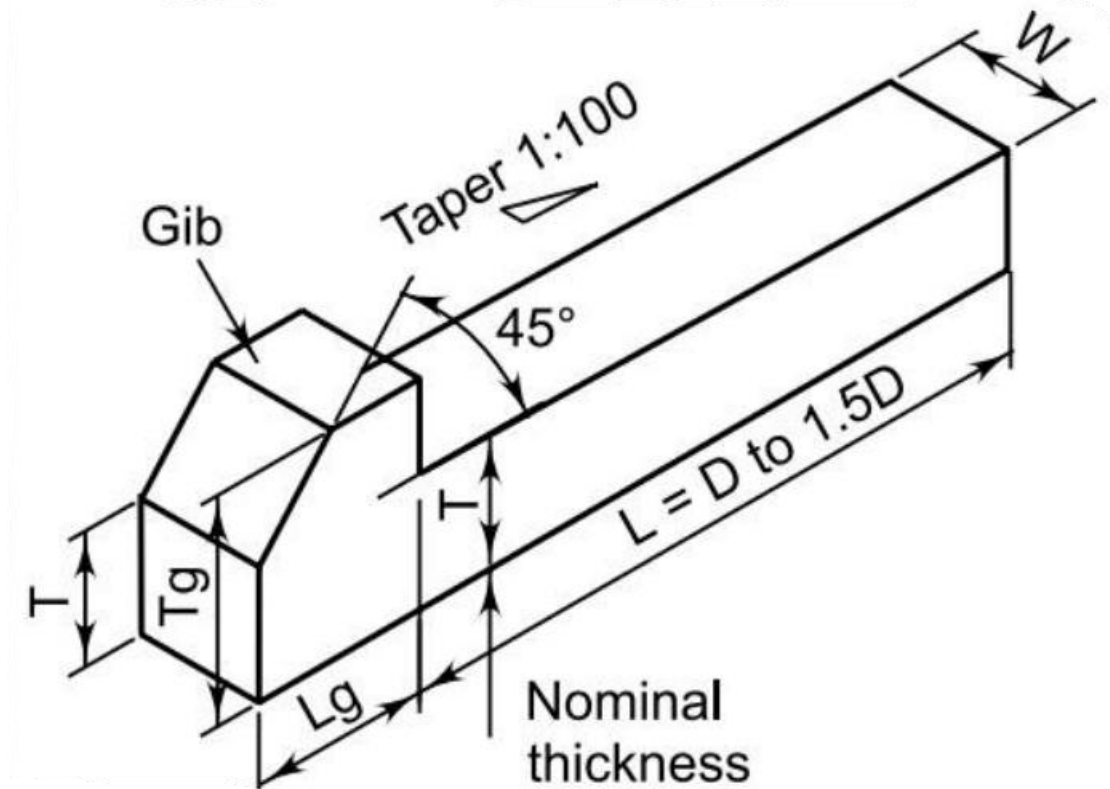
Width of key (W) = $0.25D + 2$ mm

Thickness of key (T) = $0.2D + 2$ mm

Thickness of the gib head (T_g) = $0.288D$

Length of gib head (L_g) = $0.2475D$

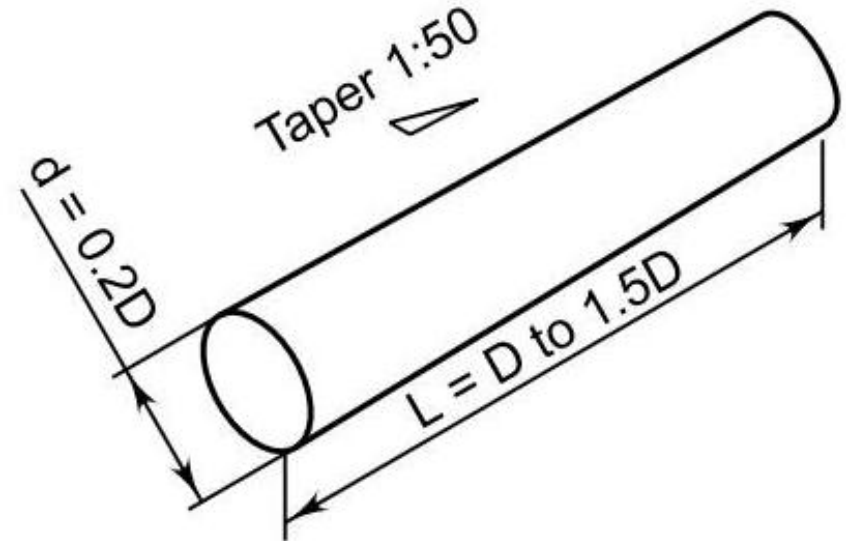
Length of key excluding gib (L) = D to $1.5D$



A Gib Head Key

D) Round Key

It is of circular cross-section of diameter $0.2D$ with a taper of 1:50 on diameter along its length. It is driven in a hole which is half in the shaft and half in the hub.



Parallel Keys

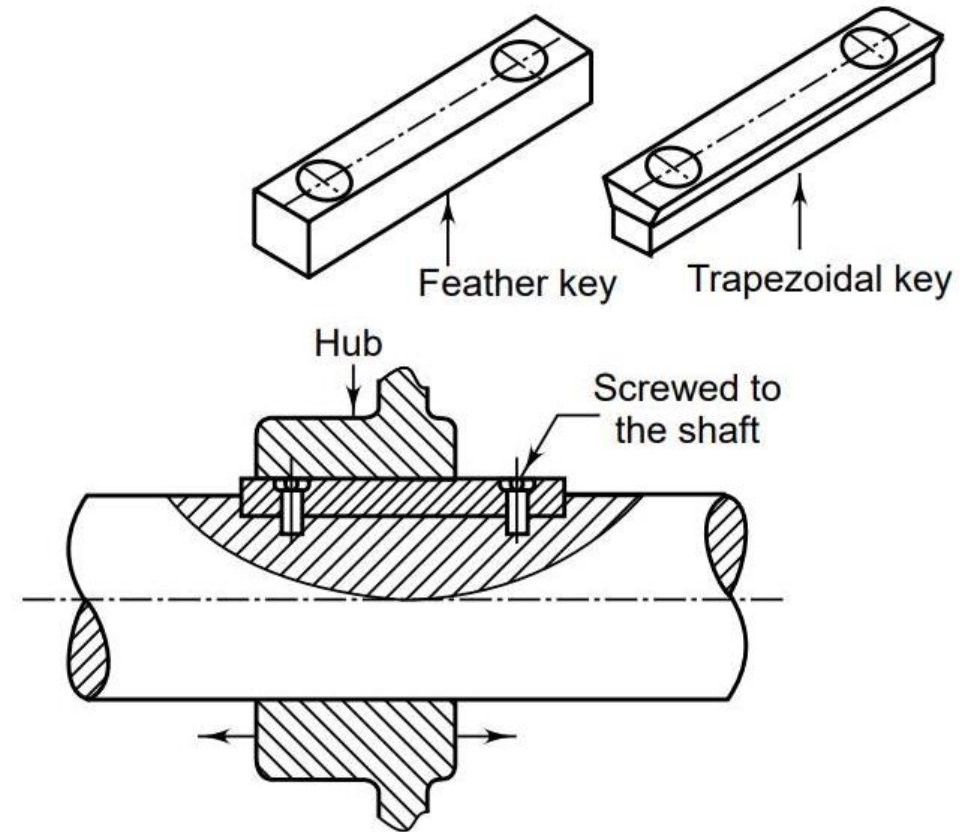
There is no taper for these keys hence thickness does not change.

Movement of key is restricted by external methods with screws, etc.

Parallel Keys - Types

A) Feather Key

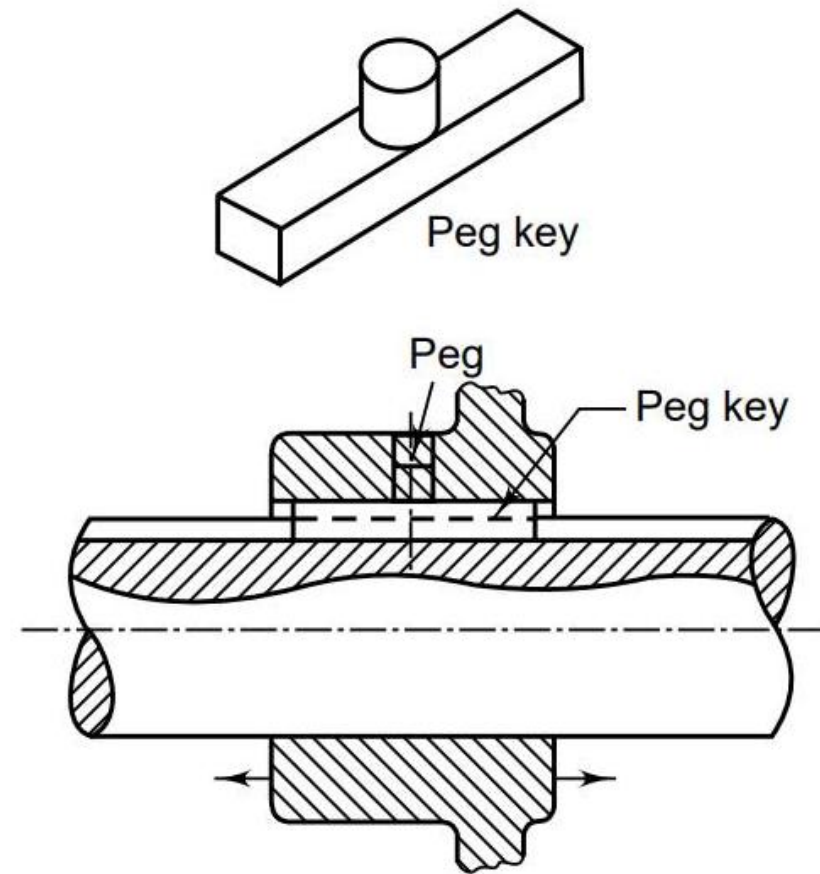
Sometimes the key is screwed on the shaft to permit axial sliding of the hub. Size of the keyway is made such that the hub can slide freely over the key. The cross-section of the key can be rectangular or trapezoidal.



Parallel Keys - Types

B) Peg Key

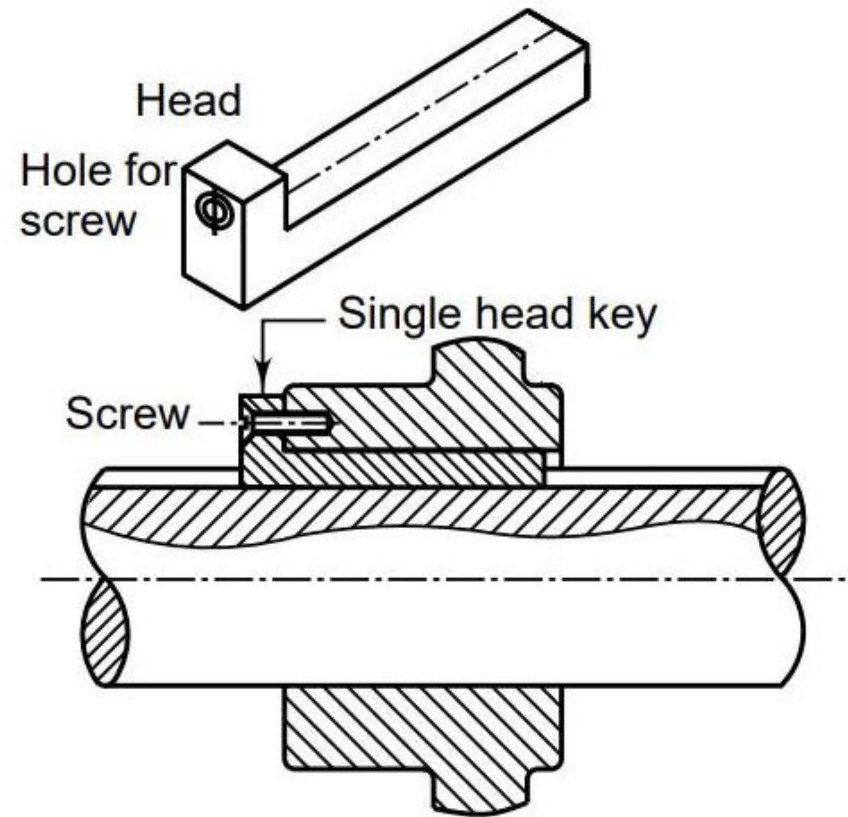
Sometimes a peg is provided on top surface of the key of same diameter as the width of key. This fits in the hole of the hub. The key can slide in the keyway of the shaft along with hub.



Parallel Keys - Types

C) Single Head Screw Key

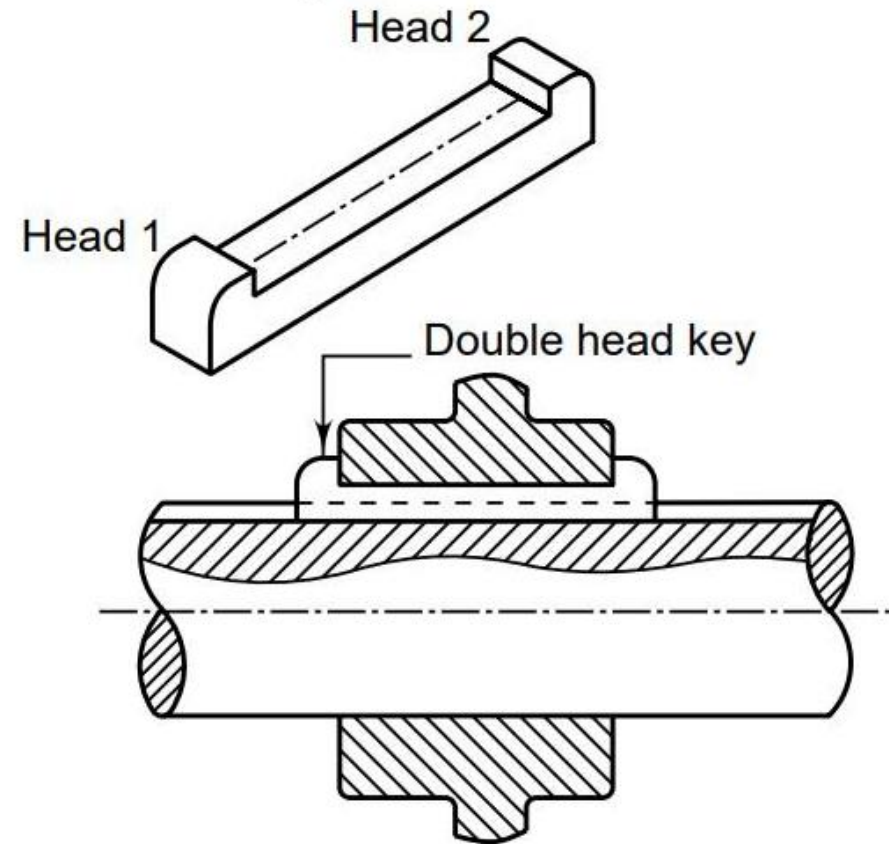
The key is fixed with a screw to the hub.



Parallel Keys - Types

D) Double Head Key

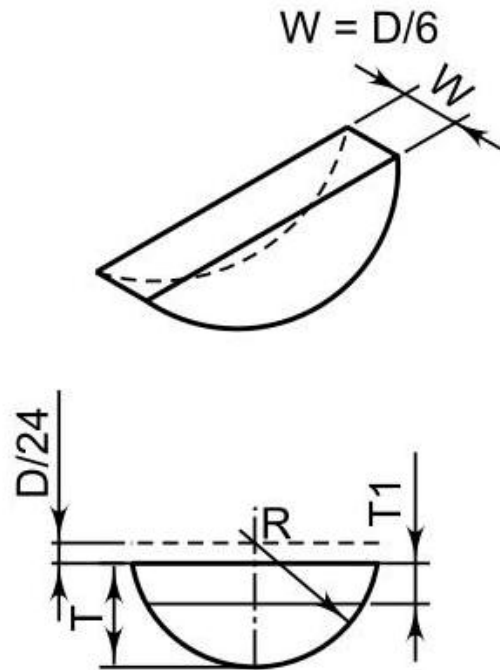
A double head key needs no external device to fix the key. It has two heads having a gap equal to hub length and therefore once fixed, cannot come out from any side.



Woodruff Key

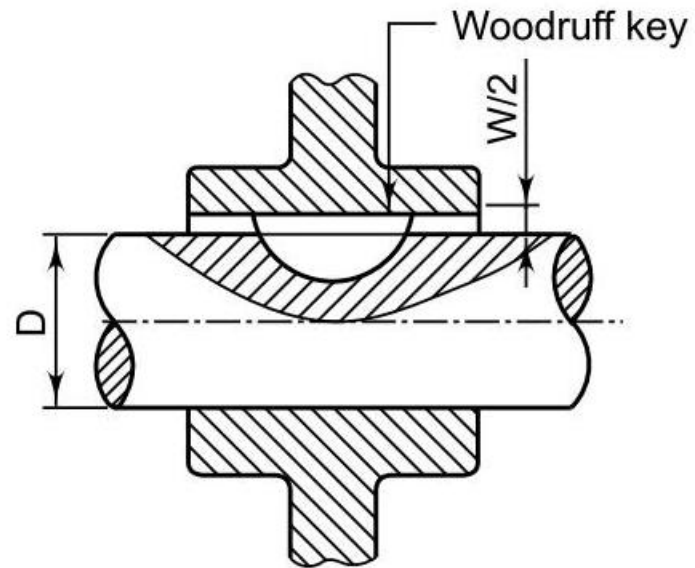
It is in the form of a segment of a circular disk of uniform width (W) equal to $D/6$. Radius of the segment (R) is kept $D/3$ and its thickness at middle (T) is $1.75W$. Its bottom end fits into the semi-circular cavity of the shaft and the top face fits into the parallel keyway of the hub. The key can rotate because of the circular base and hence it can be used for hub having tapered hole also.

Woodruff Key

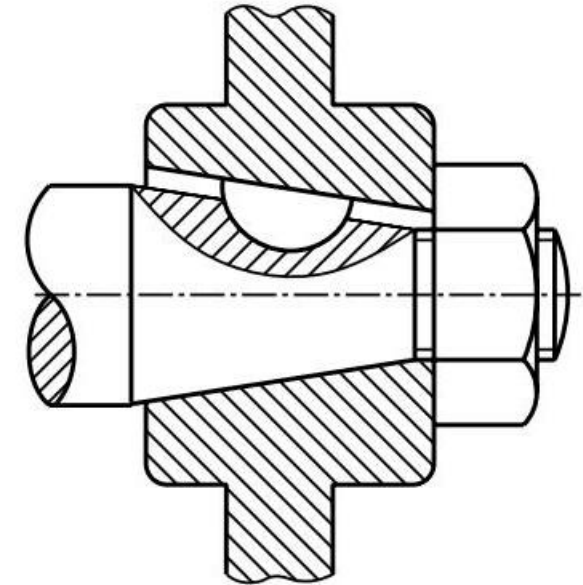


$$T = 1.75 W = 7D/24$$
$$T1 = W/2 = D/12$$
$$R = D/3$$

(A) Proportions of woodruff key shaft



(B) Key on cylindrical shaft



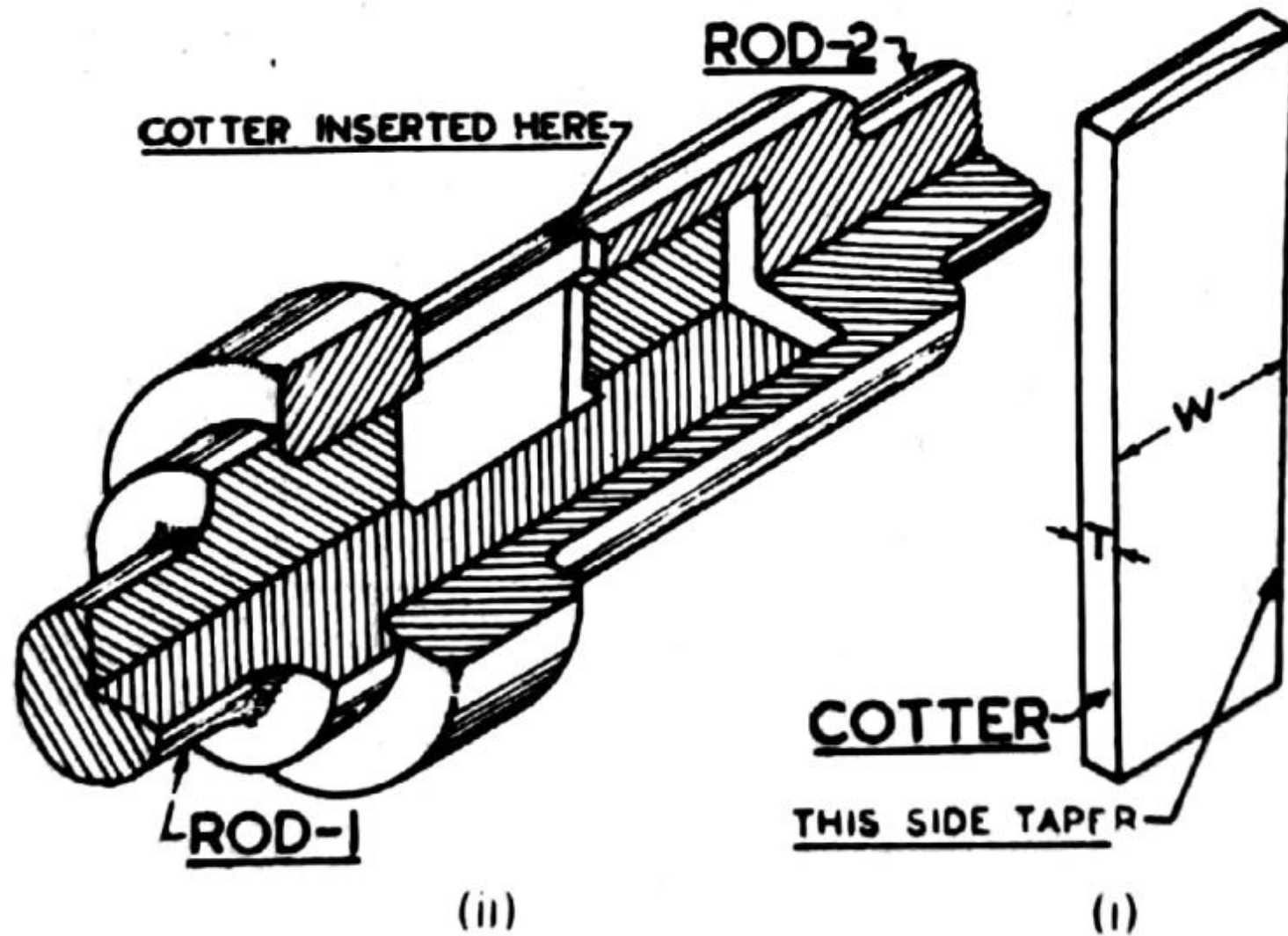
(C) Key on tapered shaft

COTTER JOINTS

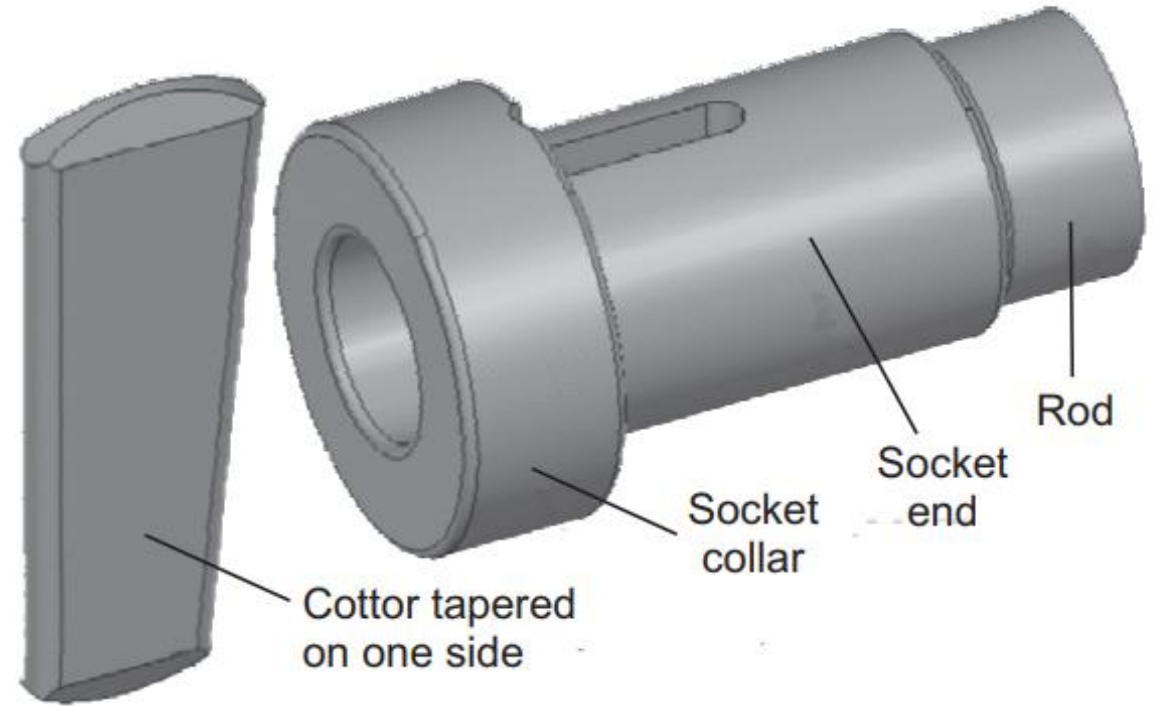
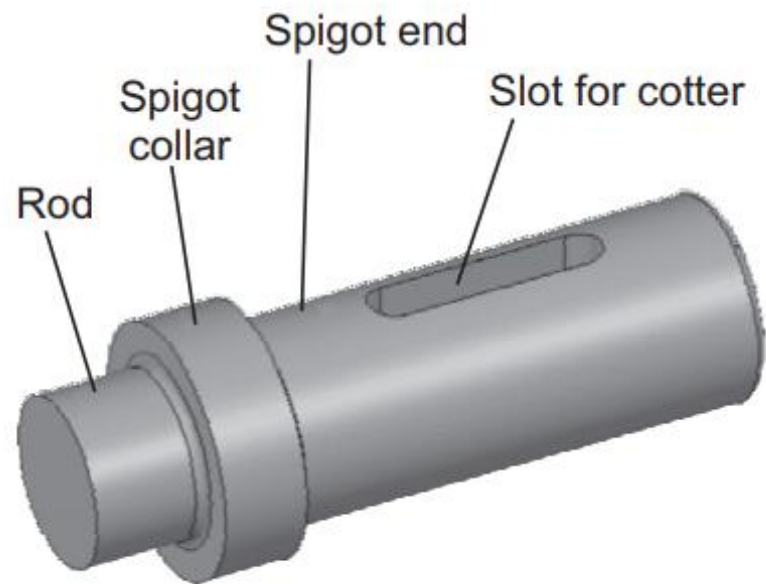
Riveted and welded joints are permanent joints. For applications where the machine elements are to be detached whenever desired, temporary joints are used. Threaded fasteners are one type of temporary joints. Cotter joints are used to join two rods having tensile or compressive loads. They cannot be used if the members rotate.

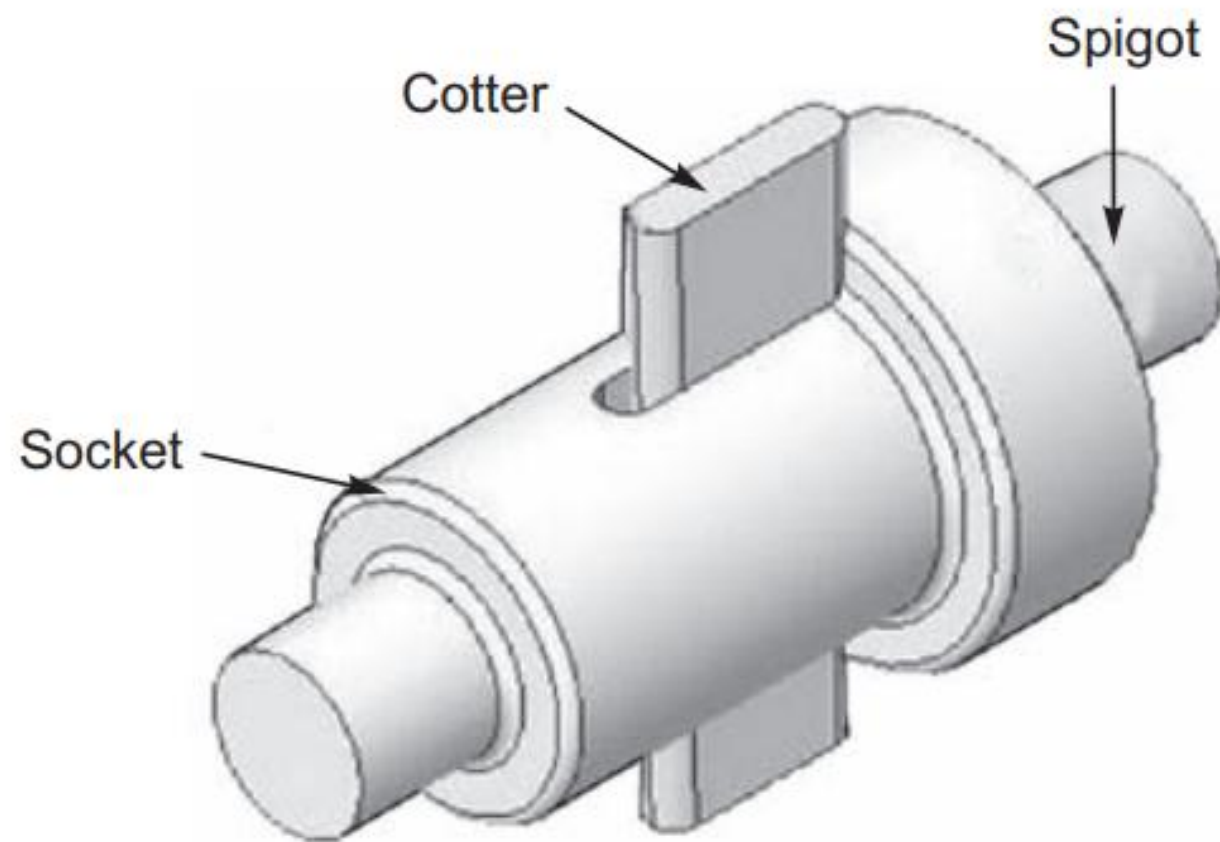
Cotter is a tapered element with rectangular cross-section of width $1.3D$ (D is the shaft diameter), thickness $0.3D$ and length $4D$. One edge is straight while the other has a taper on width of the order of 1:30 to 1:100. Sometimes the long edges are rounded. The thickness remains constant. Top and bottom edges are curved with radius equal to D , and this curved portion is tapered for thickness at 60° .

Socket and spigot joint

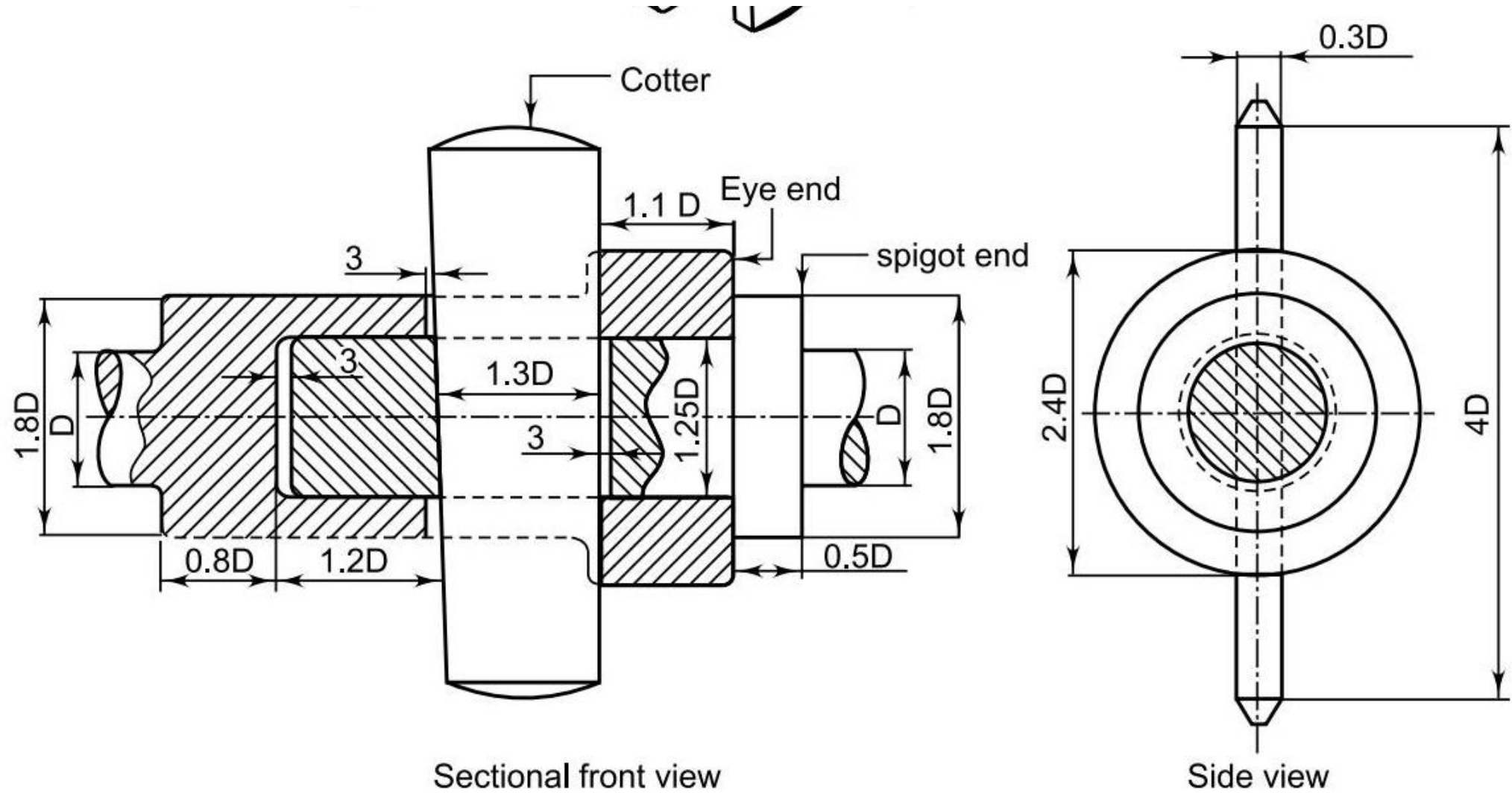


Socket and spigot joint

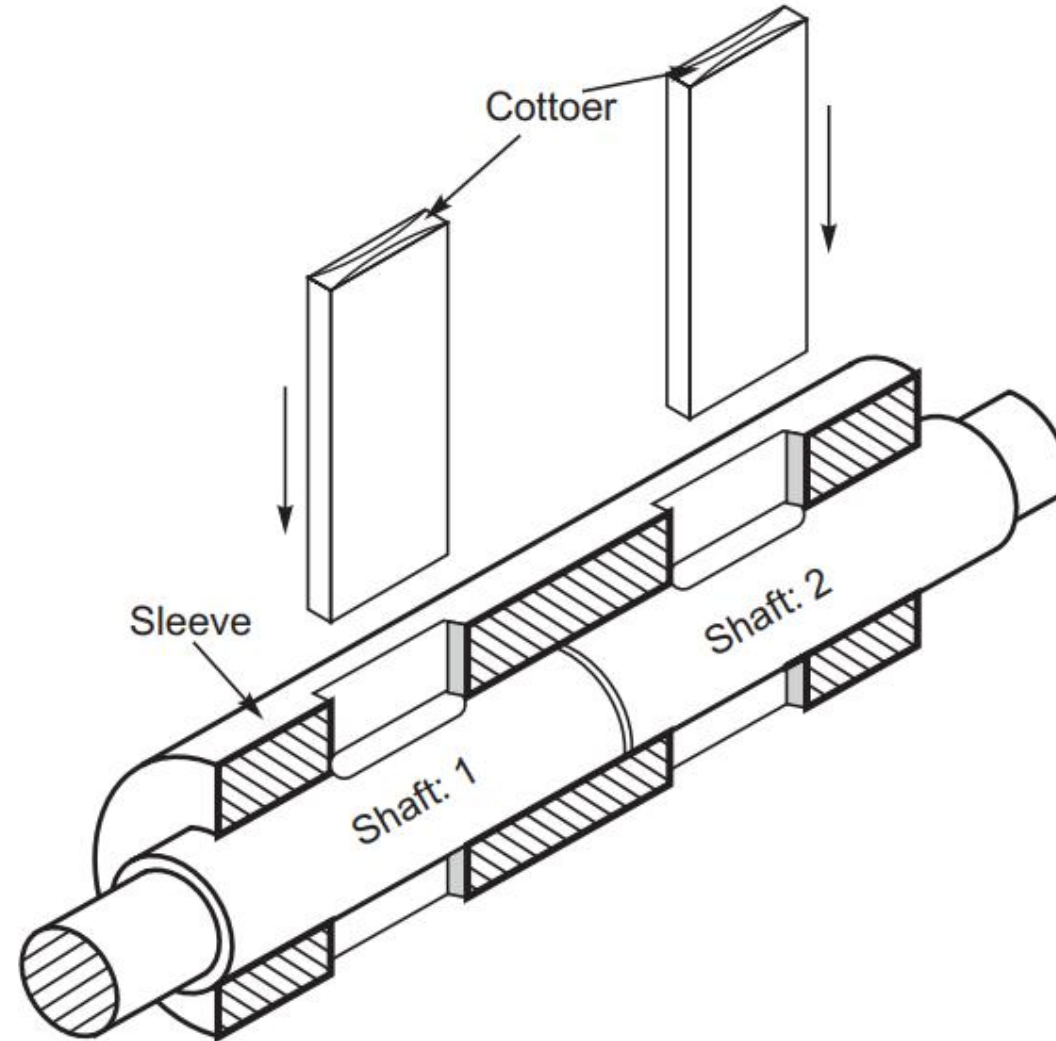




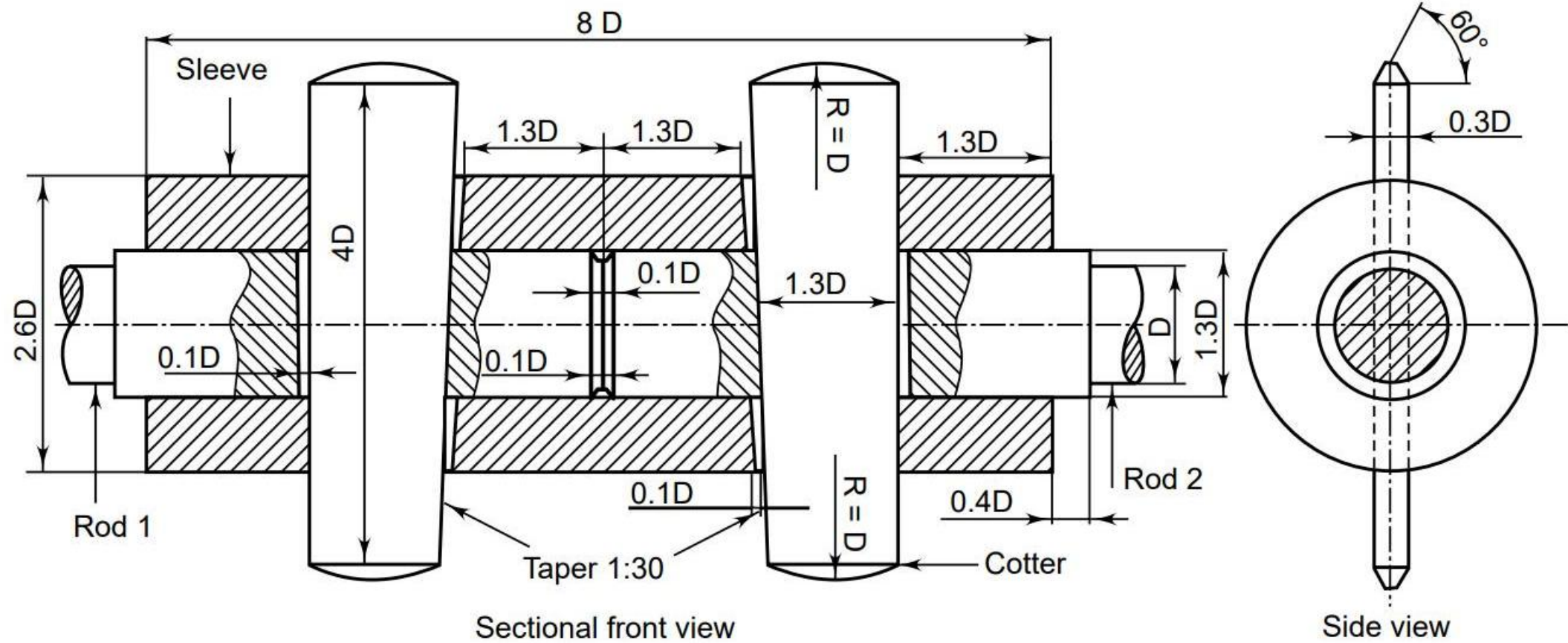
Socket and spigot joint



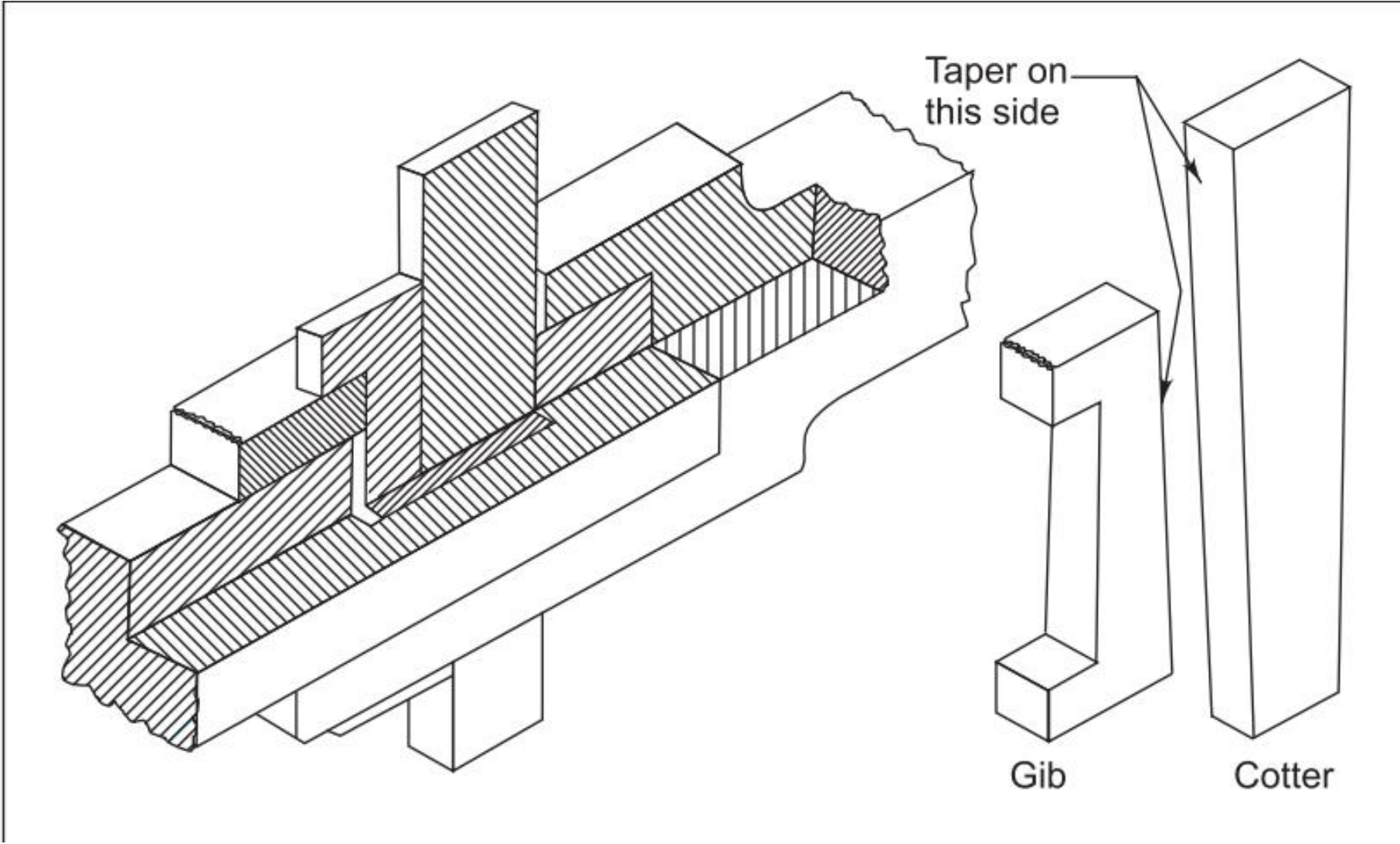
Cotter Joint with Sleeve

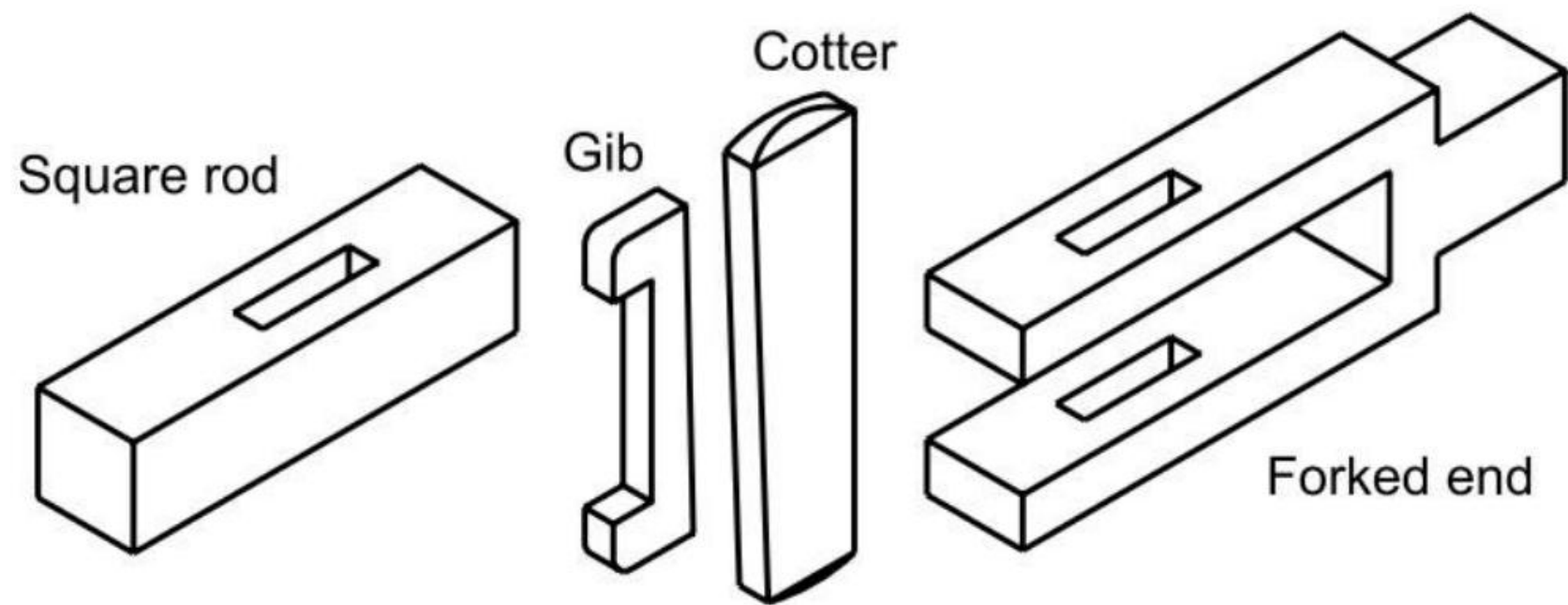


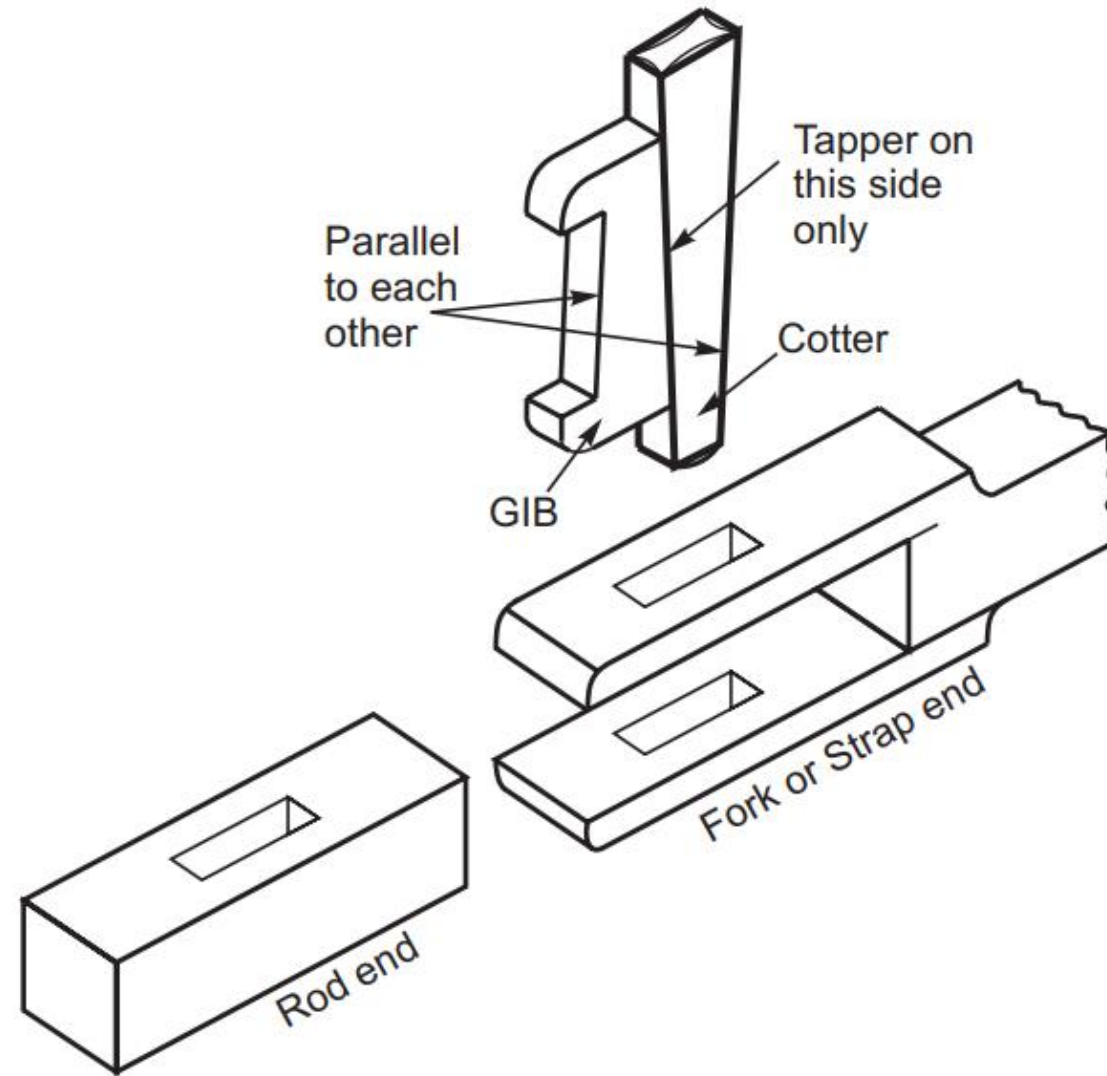
Cotter Joint with Sleeve



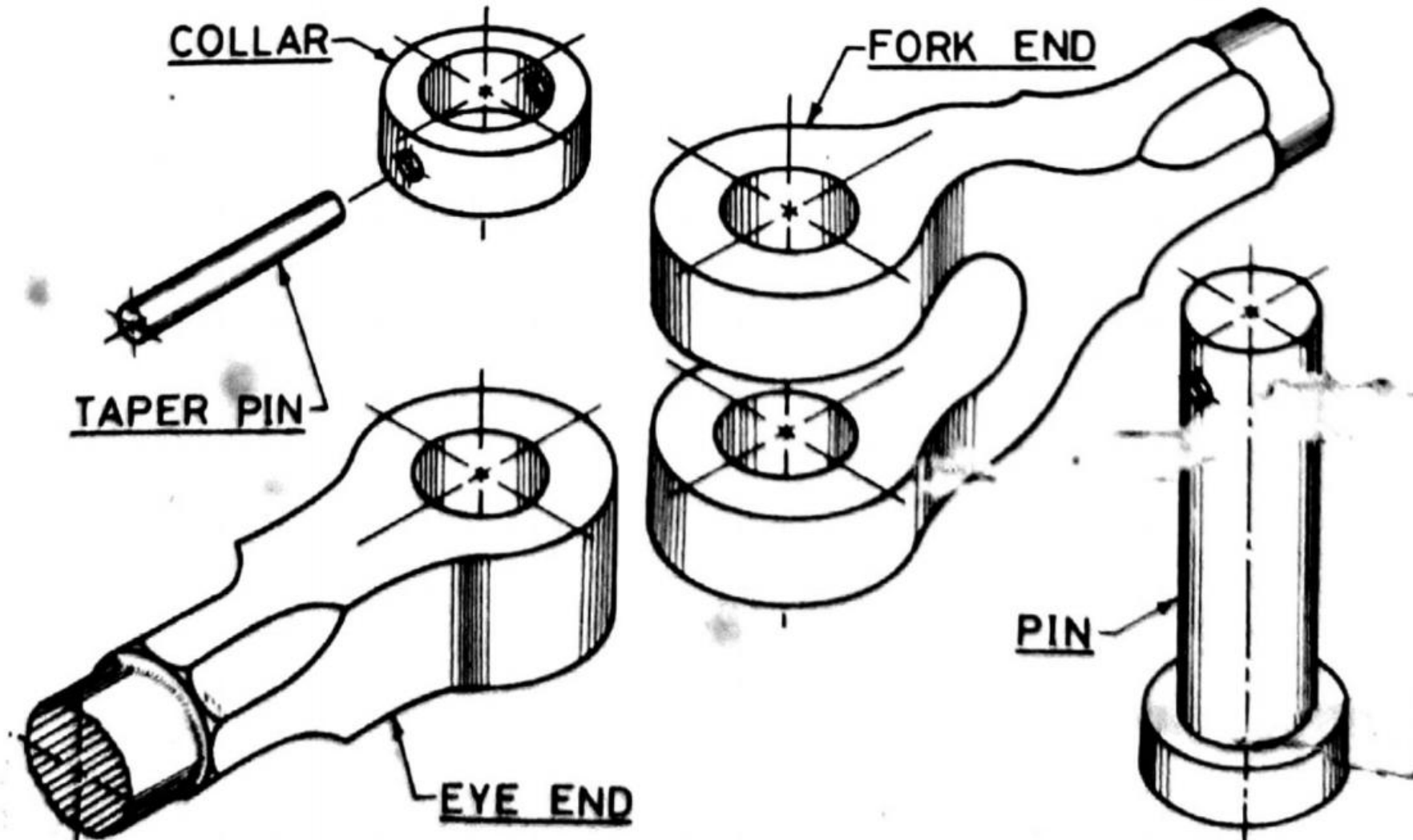
Strap joint







Pin-joint or Knuckle joint



Front view

Top view

Application Of Cotter Joint

Automotive Industry: It is used in automotive applications, such as connecting tie rods to steering arms, connecting push rods to rocker arms in the engine, or joining various linkage components.

Construction Machinery: These are employed in construction equipment, such as connecting control arms, levers, and linkages in excavators, loaders, and cranes.

Agricultural Machinery: It is find use in agricultural equipment for connecting components like linkage arms, control levers, and connecting rods in tractors, harvesters, and other farming machinery.

Industrial Machinery: It is utilized in various industrial machines, such as presses, milling machines, and lathes, for connecting drive shafts, cranks, and other rotating components.

Application Of Cotter Joint

Mechanical Engineering: It is commonly used in mechanical engineering projects for connecting rods, shafts, and spindles, particularly in low-speed or static applications.

Marine and Shipbuilding: It is employed in marine applications, such as connecting control rods, levers, and linkages in ship propulsion systems, steering mechanisms, and control systems.

Railway Industry: It is used in rail transport for connecting various components like control rods, levers, and linkages in locomotives and railcar systems.

They provide a simple and effective means of connecting components, allowing for the transmission of motion and torque in a wide range of mechanical systems.